DIAL Ecosystem — Product Requirement Document

Find it. Buy it. Get it done.
Last updated: 2026-08-07 – v1.0.0

1. Vision

Why does this thing deserve to exist?

A Zimbabwe where every vehicle owner can find a genuine part, book a vetted professional, and manage a vehicle's entire lifecycle from one trusted platform — with pricing, workmanship, and delivery all backed by DIAL.

2. Problem Statement

What pain are we relieving, and for whom?

- **Vehicle owners** face fragmented, unverified suppliers, uncertain stock, and counterfeit or wrongly-fitted parts at inconsistent prices with no recourse when something goes wrong. Left unaddressed, they keep absorbing the cost of bad parts and guesswork repairs.
- **Customers needing a mechanic, electrician, or other artisan** rely on informal referrals with no evidence of competence, punctuality, or workmanship. Without an alternative, they keep gambling on word-of-mouth with no protection when work goes badly.
- **Fleet operators** manage multiple vehicles with no consolidated view of maintenance, lifecycle cost, or licence/insurance expiry, tracked (if at all) across spreadsheets and driver reports. Left unaddressed, this keeps producing avoidable downtime and admin overhead.

- **Suppliers and technicians** have stock or skills but no reliable channel to reach demand, get paid securely, or build a portable reputation. Without a platform, they stay capped by inconsistent walk-in trade.

3. Goals & Success Metrics

How will we know we're winning?

Goal	KPI
Prove the parts marketplace in Harare	GMV and Fill Rate (% of part requests fulfilled on-platform)
Prove technician trust and liquidity	Booking conversion rate and on-platform completion rate
Sustainable unit economics	Take rate and contribution margin per transaction
Retention and repeat use	LTV:CAC ratio and 90-day repeat rate

Feature-specific health metrics to track alongside these: **stock-data freshness** (time since last supplier update, vs. supplier-confirmation failure rate), **sourcing-request conversion** (quoted → deposit paid → fulfilled), and **AI valuation accuracy** (variance between the AI's estimate and the technician's final on-site quote).

4. Personas & User Stories

Who are we building for, and what do they want?

> **Persona – Vehicle Owner Tendai**: 34, drives a 2015 Toyota Hilux for work, was once sold a counterfeit brake pad and now double-checks everything.

> **Persona – Fleet Manager Rudo**: 41, runs a 15-vehicle delivery fleet, currently tracks maintenance across a shared spreadsheet.

> **Persona – Supplier Panashe**: 38, owns a spares shop in Graniteside with both walk-in and phone-order trade, wants more customers without exposing pricing to competitors.

> **Persona – Technician Farai**: 29, auto electrician, wants steady bookings and reliable payment without chasing customers for cash.

Dial a Spare

- **US-1**: *As a* vehicle owner, *I want* to search by my vehicle's make/model/year, *so that* I only see parts that fit.
- **US-2**: *As a* customer, *I want* to compare anonymous supplier offers by brand, warranty, and delivery time, *so that* I can choose without bias toward one supplier.
- **US-3**: *As a* supplier, *I want* to upload my stock via a CSV/Excel file, *so that* my inventory appears in the catalogue without manual listing.
- **US-4**: *As a* supplier, *I want* a nudge when my listed stock looks stale, *so that* I can refresh it before overselling.
- **US-5**: *As a* customer, *I want* to request sourcing for a part I can't find, by submitting my vehicle details and a photo, *so that* I can still get rare parts.
- **US-6**: *As a* customer, *I want* to see the sourcing deposit upfront, *so that* I know my risk before DIAL starts searching.

Dial a Tech

- **US-7**: *As a* customer, *I want* to book a vetted technician by trade and location, *so that* I can trust who's coming to my vehicle.
- **US-8**: *As a* customer, *I want* to attach photos to my problem description, *so that* I get a more accurate deposit and call-out estimate before booking.
- **US-9**: *As a* technician, *I want* the AI estimate as a starting point I can adjust on-site, *so that* I'm not bound to a number the model got wrong.

Dial Fleet / Vehicle Hub / Dial Care

- **US-10**: *As a* fleet manager, *I want* a consolidated dashboard across all my vehicles, *so that* I can plan budgets and cut downtime.
- **US-11**: *As a* customer, *I want* to add insurance to my Dial Care plan, *so that* I can manage protection and maintenance in one place.
- **US-12**: *As a* customer needing a tow, *I want* DIAL to dispatch a partner operator, *so that* I get help quickly.

5. Scope

Draw the fence so the project doesn't balloon.

5.1 MVP (v1.0)

- Dial a Spare: catalogue search, anonymous multi-supplier offers, checkout, delivery/collection, returns
- `[MVP]` Supplier stock import via CSV/Excel, with unmatched rows manually reviewed and the existing supplier-confirmation step as the stock-accuracy safety net
- `[MVP]` Non-catalogue sourcing requests (photo + vehicle details, ops-reviewed quote, deposit-gated)
- Dial a Tech: technician profiles, booking, matching, ratings
- `[MVP]` Photo + description upload at booking, valued by the ops team manually (AI not yet in the loop)
- Vehicle Hub: service history, reminders
- Core payment ledger, dispute handling
- Harare only; priority parts categories and trades only

5.2 Phase 2 (v1.1)

- Automated OEM-number fuzzy matching for stock mapping, shrinking the manual review queue
- WhatsApp "mark sold" shortcut for suppliers between sheet uploads
- `[Phase 2]` AI-assisted job valuation (vision + language model), always technician-adjustable

- AI-assisted first-pass photo classification for sourcing requests, to speed up ops quoting
- Dial Fleet dashboard, preventative maintenance scheduling
- Algorithmic supplier badging
- Bluetooth thermal label printing for dispatch

5.3 Phase 3 (v2.0+)

- `[Phase 3]` Dial Care insurance integration with a licensed underwriting partner
- `[Phase 3]` Dial Assist towing/roadside partnerships (e.g. Road Angels) with live dispatch
- POS/inventory API integration for high-volume suppliers (near-real-time stock sync)
- National EPC catalogue scale-up, multi-city expansion

6. Product Features & Solutions

A high-level catalogue of how we solve each problem.

Module	Solves
Dial a Spare	Fragmented, unverified parts supply
Dial a Tech	Unverified, unaccountable services
Dial Fleet	No consolidated multi-vehicle view
Vehicle Hub	No digital vehicle record
Dial Care	No structured maintenance/protection plan
Dial Assist	No reliable roadside/breakdown response

6.1 Supplier Stock Import & Catalogue Mapping (Dial a Spare)

Suppliers upload a spreadsheet with four required columns — OEM Part Number, Stock Figure, Brand, Cost (exact file spec in Appendix C). The system normalises each OEM number (strips whitespace, dashes, leading zeros) and matches it against DIAL's master catalogue, built from EPC cross-reference data. Matched rows become live, badged offers priced at supplier cost plus DIAL's margin, delivery, and payment costs, per the net-price model already defined. Unmatched rows queue for the catalogue team to map manually or seed a new master entry.

Every live offer carries the supplier ID that generated it, so when a customer buys, the order and ledger lock in that supplier for settlement regardless of any stock update that happens afterward.

Keeping stock honest when a walk-in customer buys the same unit in-shop: a spreadsheet alone can't fully solve this, so it's handled in tiers:

- **MVP** — lean on the fulfilment workflow's existing supplier-confirmation checkpoint as the safety net. The supplier must confirm within a set SLA or the order auto-cancels, refunds the customer, and DIAL re-offers from the next supplier; repeated failures count against the supplier's reliability score and badge eligibility.
- **Phase 2** — a WhatsApp "mark sold" shortcut plus scheduled reminders to re-upload the sheet, so the safety net triggers less often.
- **Phase 3** — direct POS/inventory API integration for suppliers whose own systems support it.

6.2 Non-Catalogue Sourcing Requests (Dial a Spare)

A "Can't find your part?" entry point collects vehicle make/model/year/engine/transmission, a description, and at least one photo — the same fitment fields the master catalogue already uses. This gives concrete shape to the "assisted request and quotation engine" already named as the catalogue workaround, and doubles as data that grows the master catalogue over time. The sourcing team (from Phase 2, assisted by the same vision model used for job valuation, to pre-classify the part from the photo) returns a price and lead-time estimate. Sourcing does not start until the customer pays a security deposit, protecting DIAL and suppliers from unpaid effort on requests that were never serious. Quotes carry an expiry window, given currency volatility.

(Deposit structure and refund terms if sourcing fails are open — see Section 14.)

6.3 AI-Assisted Job Valuation & Deposit Calculator (Dial a Tech)

At booking, a customer describes the problem and can attach photos — dashboard warning lights, visible damage, leaks. A vision-and-language model reviews both to produce a preliminary trade and complexity classification, converted into an estimated security deposit and a distance-based call-out fee, reusing the approximate-distance capability already planned for technician matching.

This estimate is explicitly a starting point, not a final price, consistent with the existing tiered payment model (small jobs: flat fee or deposit; larger jobs: milestone quotes; emergency work: call-out fee first). The assigned technician confirms or adjusts it after their own on-site look. A defined variance threshold between the AI estimate and the technician's on-site quote triggers a re-confirmation step with the customer before work proceeds, so nobody is bound to a number the model got wrong. Price bands need manual calibration against real Harare labour and parts costs before this is trustworthy — a Phase 1/2 activity, not a day-one given.

6.4 Dial Care Insurance & Roadside Partnerships

Dial Care plans extend to insurance products distributed on behalf of a licensed underwriting partner — DIAL distributes and services the relationship rather than underwriting risk itself, which fits the existing "insurance and authorised service partnerships" line in the plan. In Zimbabwe this distribution role is a regulated activity: IPEC (Insurance and Pensions Commission) registers insurance agents, including a **Multiple Agent** category for entities distributing on behalf of more than one insurer, alongside individual staff certificate and clearance requirements. This needs specialist local legal input before Phase 3 (see Section 14).

Dial Assist stays asset-light by partnering with existing roadside operators rather than owning a tow fleet. **Road Angels** — a Zimbabwean 24/7 roadside assistance company operating since 2010, offering towing, tyre change, fuel delivery, lockout, jump-start, and vehicle-valuation services from branches in Harare and Bulawayo — is a strong candidate: it already holds accreditation arrangements with insurance and roadside-assistance partners (including cross-border SADC coverage), which fits naturally alongside the Dial Care insurance angle above. Dispatch mirrors the Dial a Tech pattern already defined: request → nearest available partner offered the job → accept/decline → live status shared with the customer. Whether the partnership is exclusive or one of several parallel operators is an open question (Section 14).

7. Architecture & Tech Stack

What tools power the solution, and why are they a good fit?

The existing plan favours managed infrastructure and a small team's ability to ship fast:

- **Mobile:** Native Android via Kotlin + Jetpack Compose — performance, animation quality, and deep hardware access (camera for job/part photos, Bluetooth for dispatch printing).
- **Backend & DB:** Supabase (managed Postgres) for state machines, real-time sync, auth, storage, and edge functions.
- **Web:** React/Next.js for the customer web marketplace, supplier portal, and admin portal.
- **Hardware:** Bluetooth thermal printing for dispatch labels; edge-device telemetry for fleet monitoring.

Additions to support the four new capabilities:

- **Spreadsheet ingestion:** a JS-based parser (e.g. SheetJS) inside a Supabase edge function to read supplier `.csv` / `.xlsx` uploads — the stack is already TypeScript end to end, so this drops in cleanly.
- **Fuzzy matching:** Postgres's own `pg_trgm` extension for trigram similarity on OEM numbers and descriptions — worth trying before reaching for a separate matching service, since the catalogue already lives in Postgres.
- **Vision-language model:** a pluggable multimodal API, shared by sourcing-request triage (6.2) and job valuation (6.3). Vendor selection and accuracy evaluation against real Harare jobs is a Phase 2 workstream, not assumed here.
- **Distance/geolocation:** a maps distance-matrix API for call-out fee calculation and technician proximity.
- **Payments:** provider-neutral adapters extended for Zimbabwe's mobile-money rails (Section 9), alongside card/bank rails.

Key components at a glance:

- **Front end:** Next.js (web), Kotlin/Compose (Android)
- **Back end:** Supabase (Postgres, edge functions, storage, auth)
- **Matching:** Postgres `pg_trgm` + manual review queue
- **AI:** pluggable vision-language API (vendor TBD)
- **Infra diagram:** `docs/arch-diagram.png` (carried forward, needs updating with new services)

8. High-Level Code Contracts

Give devs (and AI) the critical interfaces up front.

```
// src/api/types/SupplierStockRow.ts
export interface SupplierStockRow {
  supplierId: string
  oemPartNumber: string
  stockFigure: number
  brand: string
  cost: number
  matchedMasterProductId?: string
```

```
  matchStatus: 'matched' | 'unmatched' | 'pending_review'
  uploadedAt: Date
}

// src/api/types/SourcingRequest.ts
export interface SourcingRequest {
  id: string
  customerId: string
  vehicle: { make: string; model: string; year: number;
engineCode?: string }
  description: string
  photoUrl: string
  status: 'submitted' | 'quoted' | 'deposit_paid' |
'sourcing' | 'fulfilled' | 'cancelled'
  quotedPrice?: number
  depositAmount?: number
  quoteExpiresAt?: Date
}

// src/api/types/JobValuation.ts
export interface JobValuationRequest {
  jobId: string
  description: string
  photoUrls: string[]
  customerLocation: { lat: number; lng: number }
}

export interface JobValuationEstimate {
  jobId: string
  estimatedTrade: string
  estimatedComplexity: 'low' | 'medium' | 'high'
  estimatedDepositAmount: number
  calloutFee: number
  distanceKm: number
  isPreliminary: true // always surfaced to the customer
as "confirmed on-site"
}
```

9. Third-Party Integrations / APIs

A quick ledger of external systems we depend on.

System	Purpose	Notes
Paynow	Card, ZimSwitch, and mobile-money (EcoCash/OneMoney/TeleCash) checkout and settlement	Most established Zimbabwean gateway, with SDKs matching a TS/Node stack; evaluate Pesepay/DPO as alternatives for API ergonomics
EPC data source	Master catalogue cross-referencing	Per existing plan
Vision-language API	Photo classification for sourcing requests and job valuation	Vendor TBD, Phase 2
Maps/Distance API	Call-out fee calculation, technician proximity	—
IPEC-registered insurance underwriter	Dial Care insurance products	DIAL likely registers as a Multiple Agent (Section 14)
Road Angels (or equivalent)	Dial Assist towing/roadside dispatch	Candidate partner already accredited with insurers
WhatsApp Business API	Flows channel, supplier "mark sold" shortcut	Additional channel, not source of truth

System	Purpose	Notes
Bluetooth thermal printer SDK	Dispatch label printing	Per existing plan

10. Pricing & Monetisation

How does this venture pay for lunch?

Business Unit	Primary Revenue	Additional Revenue
Dial a Spare	Margin / supplier commission	Delivery, promoted listings, sourcing-request fee
Dial a Tech	Completed-job commission	Booking fee, protection/admin fee
Dial Fleet / Care	Monthly membership / SaaS	Insurance distribution commission, SLA prioritisation
Vehicle Hub	Premium features	Cross-sell and retention value
Dial Assist	Partner referral commission per dispatch	—

11. Deployment & Environments

Where does the code live, and how does it move from laptop to internet?

Separate Supabase projects per environment (dev/staging/prod) with migration-based schema changes; Next.js web deploys via Vercel (or equivalent) with instant rollback; CI/CD via GitHub Actions running lint/type-check/test before merge. New, higher-risk features — CSV stock import and AI valuation — ship behind feature flags to a subset of suppliers/technicians before general availability.

12. Non-Functional Requirements

Quality gates that live outside feature checklists.

- **Performance:** 95% of API calls ≤ 200ms
- **Accessibility:** WCAG 2.2 AA or better
- **Compliance:** Zimbabwe's Cyber and Data Protection Act [Chapter 12:07] (2021) — DIAL processes personal data (customer, supplier, and technician records, job photos, location) as a data controller, which under the Act's licensing regulations may require registration and a designated Data Protection Officer; POTRAZ is the enforcing authority
- **Uptime:** 99.9% monthly SLA
- **Trust/transparency:** every AI-generated estimate is labelled "preliminary — confirmed on-site" wherever it's shown
- **Data retention:** job and sourcing-request photos retained only as long as needed for the transaction or dispute window, per a defined retention schedule

13. Acceptance Criteria

Binary conditions that prove a feature is "done".

> **AC-US-3** (stock import)
> *Given* a supplier uploads a correctly formatted CSV/Excel file,
> *When* the system processes it,
> *Then* matched rows go live as offers within the target processing window, and unmatched rows are flagged for catalogue-team review.

> **AC-US-5** (sourcing deposit)
> *Given* a customer has received a sourcing quote,
> *When* they pay the security deposit,

> *Then* the request status moves to "sourcing" and the sourcing team is notified immediately.

> **AC-US-8** (AI job valuation)
> *Given* a customer submits a problem description and photos at booking,
> *When* the valuation completes,
> *Then* an estimated deposit and call-out fee are shown before booking confirmation, clearly labelled as preliminary.

> **AC-Risk-B** (stock confirmation safety net)
> *Given* a supplier fails to confirm an order within the confirmation SLA,
> *When* the SLA expires,
> *Then* the order auto-cancels, the customer is refunded, and DIAL automatically re-offers the next best supplier for the same part.

14. Open Questions & Risks

A living parking lot for unknowns so nothing slips through the cracks.

- **Q-1 (Owner: Product):** Sourcing-request deposit — flat fee or percentage of estimated value, and is it credited toward the final price?
- **Q-2 (Owner: Finance):** Refund policy if DIAL can't source a requested part within the quote window — full refund, or a smaller non-refundable search fee retained?
- **Q-3 (Owner: Ops):** Exact supplier-confirmation SLA before auto-cancel, and how repeated failures affect a supplier's badge/ranking.
- **Q-4 (Owner: Product):** Variance threshold between the AI job estimate and the technician's on-site quote that should force customer re-confirmation before work starts.
- **Q-5 (Owner: Legal/Compliance):** Whether DIAL registers with IPEC as a Multiple Agent to distribute Dial Care insurance, or whether the underwriting partner carries that role and DIAL stays a pure referral channel — needs specialist Zimbabwean insurance-law advice.

- **Q-6 (Owner: BD):** Whether a Road Angels-style towing partnership is exclusive or one of several parallel partners, and who carries liability/insurance during an active tow.
- **Q-7 (Owner: Product):** The source plan describes "customer apps" (plural) but the tech stack specifies native Android only — confirm whether iOS parity is in scope, and on what timeline.
- **Risk-A:** The vision-model API has its own per-call cost and rate limits at scale — needs capacity and cost planning once real volume is known.
- **Risk-B:** Stale spreadsheet stock data causing oversells is the biggest trust risk in the CSV-import model; the confirmation-SLA safety net needs to be tight and monitored from day one, not treated as a formality.
- **Risk-C:** Cold-start AI valuation accuracy — early wrong estimates could damage trust before enough real jobs exist to calibrate the model against actual Harare costs.

15. Timeline & Owners

Turns intentions into commitments.

Phase	Scope	Owner
Foundation	App shells, Supabase setup, EPC ingestion, manual CSV stock import	Eng Lead (TBD)
Controlled MVP	Harare launch, priority categories, sourcing requests, manual job valuation	Product Lead (TBD)
Marketplace hardening	AI valuation, fuzzy matching, WhatsApp stock shortcut	Product Lead (TBD)
Ecosystem expansion	Insurance distribution, towing partnerships, POS integrations	Ops/BD Lead (TBD)

Target dates and named owners to be confirmed at kickoff.

16. Changelog

Who changed what in this very PRD.

[1.0.0] – 2026-08-07

Added

- Full DIAL Ecosystem PRD synthesised from the August 2026 master business plan
- Supplier CSV/Excel stock import and catalogue mapping (Dial a Spare)
- Non-catalogue sourcing-request flow with security deposit (Dial a Spare)
- AI-assisted job valuation and distance-based call-out fee calculator (Dial a Tech)
- Dial Care insurance distribution model and Dial Assist towing-partnership model

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Appendices

Appendix A: Indicative Revenue Matrix

Business Unit	Primary Revenue	Additional Revenue
Dial a Spare	Margin / supplier commission	Delivery, promotion, supplier tools, sourcing-request fee
Dial a Tech	Completed-job commission	Booking fee, protection/admin fee

Business Unit	Primary Revenue	Additional Revenue
Dial Fleet/Care	Monthly membership / SaaS	Premium benefits, SLA prioritisation, insurance distribution commission
Vehicle Hub	Premium features	Cross-sell and retention value
Dial Assist	—	Partner referral commission per dispatch

Appendix B: Professional Launch Checklist

Company structure, supplier contracts, technician agreements, customer terms, payment-provider agreement, tax treatment, branded workwear, verification SOPs, dispute policy, catalogue standards, security testing, operational dashboards, **plus**: supplier CSV format documentation, sourcing-request deposit/refund terms (legal), AI-valuation disclaimer language in customer ToS, IPEC insurance-distribution registration (if applicable), towing-partner SLA and liability agreement.

Appendix C: Supplier Stock Import — File Specification

Column	Field	Type	Required	Notes
A	OEM Part Number	text	Yes	Normalised (whitespace/dashes/leading zeros stripped) before matching against the master catalogue
B	Stock Figure	integer	Yes	Quantity on hand at time of upload
C	Brand	text	Yes	Manufacturer/brand; feeds catalogue mapping, not

Column	Field	Type	Required	Notes
				necessarily shown directly to the customer under the anonymous-offer model
D	Cost	decimal	Yes	Supplier's net ask price; DIAL adds margin, delivery, and payment costs on top

Accepted formats: `.csv`, `.xlsx`. Maximum file size and row count per upload: TBD with ops.